Hospital Information System Application

This project aims to develop an application simulating a Hospital Information System (HIS) for a chosen **department**:

- Radiology
- Emergency
- Hemodialysis
- Cardiology
- Operation Rooms
- ICU Rooms
- Ophthalmology Department
- Internal Medicine Department
- Surgery Department
- Obstetrics and Gynecology Department

Essential Functionalities:

- Home page for visitors.
- Different users can use the application, and each category can do different staff
- (e.g. Doctors, Patients,nurses,empkoyees, etc.)
- User registration and login for secure access
- Each user has his own profile (Think about the doctor and patient profiles)
- Static file serving and file uploads (e.g., patient scans)
- Appointments model between doctors and patients.
- Contact forms for user inquiries and requests
- Statistical analysis dashboard for administrative users (admin dashboar

Consider the following set of requirements for a hospital database that is used to keep track of patients' records

- The hospital keeps track of each patient's name, patient-number, social security number, address and phone, birthdate, sex, medical history and medical status (blood pressure, heart rate and temperature). Both of the social security number and patient-number have unique values for each patient. We keep track of the date he was admitted to the department.
- Each department is described by a name, department-code, and a chairman (a doctor). We keep track of the date he has started the department supervision. Both name and code have unique values for each department. A department may have several locations in the same time.

Each doctor is described by his personal data, (name, sex, birth-date, social security umber, major scientific area and degree). Each one joins only one department in a specific date. Each doctor examines and checks one patient or more on regular basis. Many doctors may investigate one patient in the same time, each doctor should keep track on the number of hours per week spent on each patient.

- A **doctor** may write a **prescription** of the necessary medication (s) for each **patient**. The prescription date is recorded. The doctor should give the **directions** for each medication (how many times per day and dose) in the prescription. We keep track of both the start and end dates of getting each medication. Each may write several prescriptions for many patients. Many doctors may write many prescriptions for one patient.

ALL attributes written in Bold have unique values

a-Design and draw an ER diagram for that schema.

b-Map the ER schema into relational schema.

Requirements

1. Model patients, with complete info
2. Model hospitals including regular rooms, . etc
3. Model doctor info
4. Model work relationship between doctor and hospital
5. Model treatment relationship between doctor and patient
6. Model geo locations, (to find nearest place)
7. Allow patients to register/pay for doctor appointment

8. Allow patients to cancel/get refund for doctor appointment

9. Allow doctor to reserve different rooms for patient treatment

10. Generate reports for doctor appointments, room allocation …